Distributed Model Predictive Control for Virtually Coupled Heavy-Haul Trains

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***Abstract*—In this paper, a virtual coupling control strategy for heavy-haul trains (HHTs) based on a multi-mass-point model is proposed, aiming to further enhance train capacity and operational efficiency. Compared to traditional single-mass-point model, the proposed multi-mass-point model provides a more accurate representation of HHT dynamics. A optimization problem is formulated to minimize speed tracking error and energy consumption, and individual controllers are designed for the lead HHT and each following HHT within the framework of distributed model predictive control (DMPC). The lead train tracks an optimal reference trajectory, while each following train tracks the speed of its preceding train and maintains a safe yet smallest possible tracking interval. Finally, the effectiveness of the method is verified by numerical simulation, demonstrating its excellent performance in speed tracking, minimizing in-train force and interval maintenance, even in the presence of random disturbances.**

***Index Terms*—heavy-haul train, virtual coupling, multi-mass-point model, model predictive control**

I. Introduction

Globally, heavy-haul transportation is critical for the transportation of bulk cargoes, such as coal and iron ore, due to its large capacity, high efficiency, low cost and low carbon emissions. Recent technological innovations in the heavy-haul train operation, particularly in automation and intelligence, have become central to ensuring safety, efficiency, and operational excellence in heavy-haul railways. In China, rapid societal and economic growth has driven a continuous increase in cargo transport volumes, outpacing the capacity of existing heavy-haul railways. Enhancing capacity can be achieved through constructing new railways, increasing the capacity of single trains, or increasing the number of HHTs. Among these, increasing train frequency is one of the most cost-saving and feasible solutions, given the current equipment level of heavy-haul trains.

In recent decades, communication-based train control (CBTC) systems have driven advancements in modern railways. The railway signaling system has evolved from fixed block signaling (FBS) to the moving block signaling (MBS) model, as shown in Fig. 1(a)(b). Unlike FBS, MBS uses on-board signaling, allowing movement authority blocks to move with trains instead of being fixed on tracks. This reduces the distance between consecutive trains to the absolute braking distance (ABD) plus a safety margin, significantly improving traffic efficiency [1]. However, this conservative approach still results in larger train tracking intervals between trains. In reality, depending on the braking rates of the two consecutive trains, the safety interval between two consecutive trains can be reduced. Moreover, a communication link between the trains can also help further minimizing this interval [2].

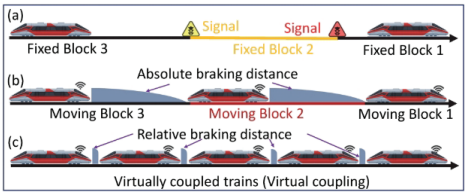


Fig. 1: Railway signaling models: (a) Fixed block; (b) moving block; and (c) virtual coupling.

Over the past several years, a new model of railway signaling communication that can further reduce the tracking interval between trains was investigated, which is called virtual coupling (VC), as shown in Fig. 1(c). The concept was first introduced in 1998 [3], and rapidly garnered significant interest from both railway professionals and academic researchers. Since then, it has been frequently referenced and innovatively adapted in numerous literatures. During recent years, there are more and more methods to realize virtual coupling, among which, the model predictive control method has been successfully implemented on numerous occasions to enhance the cooperative control performance and tracking performance of the virtual coupling trains [2], [4], [5]. [6] and [7] proposed a distributed robust model predictive control strategy. Sliding mode control was utilized in [8] and [9] to solve the formation control problem for virtually coupled trains under uncertainties. Building on these studies, [10] explored dynamic coupling and decoupling scenarios for trains. However, current research have primarily focused on high-speed trains and metros, with a

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notable absence of studies in the domain of heavy-haul trains.

In recent years, more and more researchers have dedicated their efforts to the study of virtual coupling scenarios for heavy-haul trains. [11] proposed a cooperative cruise controller for virtual train formations using multi-agent theory [12]. The above studies on virtual coupling heavy-haul trains are all based on a simple single-mass-point model, which is a key question worth pondering. Therefore, Considering that the in-train force is of great importance for heavy-haul trains, so the application of a multi-mass-point model is necessary and positive, but relevant research has yielded only a limited number of results so far, like [13] and [14]. However, [14] just employs a data-driven methodology to investigate the virtual coupling cooperative control.

The research of virtual coupling has extremely positive impact on further improving the capacity of heavy-haul trains, and meanwhile, it is realistic and necessary to describe the dynamics of heavy-haul train operation with multi-mass-point model. In this paper, we apply the multi-mass-point model that can reflect the dynamics within heavy-haul train and set up different controllers for the lead HHT and each following HHT based on distributed modeel predictive control (DMPC) respectively, which can realize the cooperative and efficient operation of vitually coupled trains.

The rest of this paper is structured as follows. Section II sets up the desired tracking interval, introduces the multi-mass-point model dynamics model of heavy-haul train and also briefly describes how the optimal reference trajectory is obtained; In section III, based on the DMPC framework, an optimization problem is designed for cooperative tracking and maintaining safe intervals between consecutive HHTs, which is then transformed into a nonlinear programming form to solve. Section IV presents simulation results to validate the effectiveness of the proposed method, even in the presence of random disturbances, and Section V provides conclusions and future perspectives.

II. Problem Formulation

This section primarily describes the cooperative control problem for virtually coupled trains to be addressed. Consider virtually coupled train system consisting of m HHTs, which is illustrated in Fig. 2.

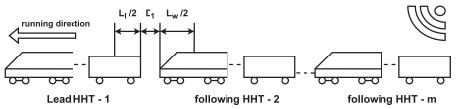


Fig. 2: A virtually coupled train system consisting of m HHTs

A. Multi-Mass-Point Model Based HHT Dynamics

Assume each HHT i is a long formation heavy-haul train hauled by multiple locomotives as shown in Fig. 3. In this configuration, Each vehicle, which could be a locomotive or a wagon, is treated as a mass point, and the coupler used for connecting adjacent vehicles is treated as a simplified spring damper model [15]. Hence the heavy-haul train can be regarded as n mass points connected by $n-1$ couplers.

The in-train force $f_{i,j}^{in}$ between the jth and $(j+1)$th vehicle in ith HHT is expressed as,

$$\begin{aligned} f_{i,j}^{in} = k_{i,j}^{'}\left(s_{i,j} - s_{i,j+1}\right) + d_{i,j}\left(v_{i,j} - v_{i,j+1}\right), \\ i = 1, 2, \ldots, m; j = 1, 2, \ldots, n, \end{aligned} \tag{1}$$

where $k_{i,j}^{'}$ and $d_{i,j}$ represent the coefficients related to the elastic and damping effect of the jth coupler, and $s_{i,j}$ and $v_{i,j}$ represent the position and speed of the jth vehicle. From an engineering perspective, the damping effect of the couplers causes nonlinear characteristic of the train model, and it is usually ignored, i.e, $d_{i,j} = 0$.

When considering the railway, it is important to take into account the resistance caused by gradients, curves, tunnels, and other factors, expressed as $f_{i,j}^{ra}$, $f_{i,j}^{ro}$ expresses the operating resistance of each vehicle, which includes rolling resistance and aerodynamic drag.

Based on the analysis provided above, the dynamics of HHT i $(i = 1, 2, ..., m)$ can be described by the following multi-mass-point model:

$$\begin{cases} \dot{s}_{i,1} = v_{i,1}, \\ \dot{v}_{i,1} = \left(u_{i,1} - f_{i,1}^{in} - f_{i,1}^{ro} - f_{i,1}^{ra}\right)/m_{i,1}, \\ \dot{s}_{i,j} = v_{i,j}, \quad j = 2, 3, \ldots, n-1 \\ \dot{v}_{i,j} = \left(u_{i,j} + f_{i,j-1}^{in} - f_{i,j}^{in} - f_{i,j}^{ro} - f_{i,j}^{ra}\right)/m_{i,j}, \\ \qquad j = 2, 3, \ldots, n-1 \\ \dot{s}_{i,n} = v_{i,n}, \\ \dot{v}_{i,n} = \left(u_{i,n} + f_{i,n-1}^{in} - f_{i,n}^{ro} - f_{i,n}^{ra}\right)/m_{i,n}, \end{cases} \tag{2}$$

where $m_{i,j}$ denotes the mass of the jth vehicle in kg, and $u_{i,j}$ is the output force of the jth vehicle in kN. For jth locomotives, $u_{i,j} > 0$ indicates the locomotive is in state of traction, while $u_{i,j} < 0$ indicates the locomotive is in braking state. For jth wagons, $u_{i,j} <= 0$, which means that wagons can only provide braking force.

According to [15], the operating resistance $f_{i,j}^{ro}$ of the jth vehicle is expressed as

$$f_{i,j}^{ro} = m_{i,j}(A_j + B_j v_{i,j} + C_j v_{i,j}^2), \tag{3}$$

where A_j, $B_j$and C_j are coefficients related to the rolling resistance and the aerodynamic drag of the jth vehicle, and are determined by experiments.

The additional resistance $f_{i,j}^{ra}$ of the jth vehicle is determined by the gradient and curvature of the train's running railway line, which is expressed by the following equation:

$$f_{i,j}^{ra} = (g \sin\theta_{i,j} + \frac{600}{R_{i,j}})m_{i,j}, \tag{4}$$

where $\theta_{i,j}$ represents the slope angle and $R_{i,j}$ represents the curve radius at the position of jth vehicle in ith HHT. At this point, we have established a model for the dynamics of heavy-haul train operation.

Eq. (2) can be written in the following compact form as

$$\dot{x_i} = f_t(x_i, u_i), \tag{5}$$

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Fig. 3: An example of heavy-haul train i.

where the state is $x_i = [v_{i,1}, ..., v_{i,n}, s_{i,1}, ..., s_{i,n}]^T \in \mathbb{R}^{2n}$, the control input is $u_i = [u_{i,1}, ..., u_{i,n}]^T \in \mathbb{R}^n$.

To apply model predictive control, we discretize the dynamics (5) into the following discrete-time equation:

$$x_i(k+1) = f(x_i(k), u_i(k)). \tag{6}$$

where k denotes the time instant, Δt the sampling time interval, and $f(x_i(k), u_i(k)) = x_i(k) + \Delta t \cdot f_t(x_i(k), u_i(k))$.

B. Control Objective

As shown in Fig. 2, the tracking interval between the ith and $i+1$th HHT is expressed as

$$D_i = s_{i,n} - s_{i+1,1} - \frac{L_l}{2} - \frac{L_w}{2}, i = 1, 2, ..., m, \tag{7}$$

where $s_{i,n}$ is the position of the last rail vehicle of lead HHT (or preceding HHT) and $s_{i+1,1}$ is the position of the first vehicle of following HHT. The length of locomotive and wagon are expressed as L_l and L_w respectively.

The control objectives can be formulated as follows:

$$D_i \to D_{des} \tag{8a}$$

$$v_i \to v_{i-1} \tag{8b}$$

$$f_{i,j}^{in} \to 0 \tag{8c}$$

where the desired distance D_{des} is used as the tracking objective for the follower to maintain a more conservative and safer distance, greater than the minimum distance D_{min}, the minimum distance D_{min} denotes the minimum interval between consecutive HHTs because of the presence of the position error. In addition, the virtual coupling HHTs are expected to achieve the same speed as preceding HHT while satisfying the minimization of the in-train force $f_{i,j}^{in}$.

Note that here we need to assume that each HHT has access to the position and speed information of all its own vehicles, and that there is a bidirectional communication link between the preceding and following HHT. In conclusion, the preparatory work for the optimal control design has been completed.

III. DMPC Based Virtual Coupling Controller

For the cooperative tracking control of virtual coupling HHTs, a distributed model predictive control approach is proposed for the real time control in this paper. Based on the receding horizon principle, at every time step, the MPC algorithm determines the optimal control inputs and state variables by solving an optimization problem over a finite prediction horizon.

To solve the optimization problem, the predictive horizon N_p and control horizon N_c are considered at time k. The notation $x_i(k+l \mid k)$ represents the state vector at time $k+l$, predicted at time k, obtained by starting from the current $x_i(k \mid k) = x_i(k) \equiv x_i(k)$. And $u_i^*(k) = [u_i(k \mid k), ..., u_i(k+N_c \mid k)]$ denotes the optimal control sequence obtained by rolling optimization.

The framework of DMPC algorithm for solving the virtual coupling problem for HHTs is shown in Fig. 4. A group of m HHTs is considered, and MPC controllers are designed for each of them, where the cost function of the lead HHT is different from that of the following HHTs.

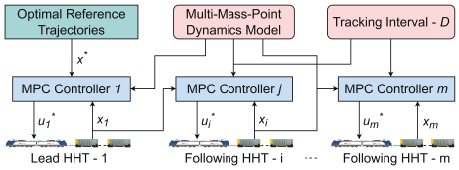


Fig. 4: DMPC framework for virtually coupled HHTs.

A. Controller Design for the Lead HHT

The control objective of the lead HHT is to track the speed profiles of the corresponding rail vehicle mass points under the constraints of speed limit and safety, and to achieve the minimum tracking interval errors while minimizing the energy consumption. Therefore, the multi-objective optimization problem for the lead HHT can be formulated as:

$$\begin{aligned}\min_{u_1} J_1 = &\sum_{l=0}^{N_p} (K_v^l \parallel v_1(k+l \mid k) - v^* \mathbf{1}_n(k+l) \parallel^2 \\ &+ K_{in}^l \sum_{j=0}^{n} \parallel s_{1,j}(k+l \mid k) - s_{1,j+1}(k+l \mid k) \parallel^2) \\ &+ \sum_{l=0}^{N_c} K_u^l \parallel u_1(k+l \mid k) \parallel^2,\end{aligned} \tag{9}$$

subject to

$$x_1(k+l+1 \mid k) = f(x_1(k+l \mid k), u_1(k+l \mid k)) \tag{10a}$$

$$-u_{min}^l(v_{1,j}) \le u_{1,j}(k+l \mid k) \le u_{max}^l(v_{1,j}), \text{ if } j \text{ is locomotive} \tag{10b}$$

$$-u_{min}^w \le u_{1,j}(k+l \mid k) \le 0, \text{if } j \text{ is wagon} \tag{10c}$$

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$$0 \le v_{1,j}(k+l \mid k) \le v^{lim} \tag{10d}$$

$$-f^{in}_{max} \le f^{in}_{1,j}(k+l \mid k) \le f^{in}_{max} \tag{10e}$$

for all $j = 1, 2, ..., n$.

The cost function (9) includes three terms, the first term is used for the lead HHT to track optimal reference speed $v^*\mathbf{1}_n$, $\mathbf{1}_n = [1, ..., 1]^T$, the second term is to minimize the in-train force, which is the relative displacement between adjacent vehicles in ith HHT, the third term is the control input, $K^l_v \ge 0$, $K^l_{in} \ge 0$ and $K^l_u \ge 0$ are weight coefficients corresponding to balance the speed tracking errors, in-train forces and control forces respectively.

In equation (10), (10a) is the system dynamic updates for the discrete-time model obtain from (6) applied to the lead HHT, (10b) and (10c) are the control force constraints for locomotives and wagons respectively, in which $-u^l_{min}(v_{1,j})$ and $u^l_{max}(v_{1,j})$ are the lower and upper bounds of the locomotive force, corresponding to the speed, and $-u^w_{min}(v_{1,j})$ is the lower bound of the wagon braking force. (10d) is related to the speedlimit of the railway and (10e) is the safety constraints related to in-train force, both are the hard constraints to ensure that HHTs can fulfill the transportation.

B. Controller Design for the Following HHT

The following HHT can receive the real-time speed and position information from the preceding HHT, the objective of which is to maintain the same speed as the preceding HHT while satisfying the minimum tracking interval. The multi-objective optimization problem for the following HHT i ($i = 2, 3, ..., m$) can be formulated as:

$$\begin{aligned}\min_{u_i} J_i = &\sum_{l=0}^{Np}(K^f_v \parallel v_i\,(k+l \mid k) - v_{i-1}\,(k+l \mid k) \parallel^2 \\ &+ K^f_D \parallel D_i\,(k+l \mid k) - D_{des}\,(k+l \mid k) \parallel^2 \\ &+ K^f_{in} \sum_{j=0}^{n} \parallel s_{i,j}(k+l \mid k) - s_{i,j+1}(k+l \mid k) \parallel^2) \\ &+ \sum_{l=0}^{Nc} K^f_u \parallel u_i\,(k+l \mid k) \parallel^2\end{aligned} \tag{11}$$

subject to

$$x_i(k+l+1 \mid k) = f(x_i(k+l \mid k), u_i(k+l \mid k)) \tag{12a}$$

$$-u^l_{min}(v_{i,j}) \le u_{i,j}(k+l \mid k) \le u^l_{max}(v_{i,j}), \quad \text{if } j \text{ is locomotive} \tag{12b}$$

$$-u^w_{min} \le u_{i,j}(k+l \mid k) \le 0, \text{if } j \text{ is wagon} \tag{12c}$$

$$0 \le v_{i,j}(k+l \mid k) \le v^{lim} \tag{12d}$$

$$-f^{in}_{max} \le f^{in}_{i,j}(k+l \mid k) \le f^{in}_{max} \tag{12e}$$

$$D_i(k+l \mid k) \ge D_{min} \tag{12f}$$

for all $j = 1, 2, ..., n$.

Compared with the controller of the lead HHT, the following HHT controller includes an additional term to minimize the tracking interval errors. $K^f_D \ge 0$ is a coefficient to penalie the output deviation from the desired interval, together with $K^f_v \ge 0$, $K^f_{in} \ge 0$ and $K^f_u \ge 0$, which has similar meaning that for the Controller of lead HHT. Eq. (12f) is used to guarantee that the interval remains within the permissible limits, and other constraints are consistent with Eq. (10).

IV. Numerical Simulation

In this section, the 653-kilometer-long Daqin Railway is used as a simulation scenario, and the simulation object will be a 20,000-ton heavy-duty train, consisting of two HXD2 electric locomotives and 214 C80B fully loaded wagons, with a total length of 2.7 kilometers, the parameters of HXD2 locomotive and C80B wagon are shown in Table I. Due to communication transmission limitations, the demand fence method [16] is applied to group a certain number of neighboring vehicles connected as rigid bodies.

TABLE I: Parameters of HXD2 locomotive and fully loaded C80B wagon

	HXD2	C80B
m_j(tons)	225	125
l(m)	37.95	12.00
$k_j(Nm^{-1})$	$78 \sim 121 \times 10^6$	$29.29 \sim 49 \times 10^6$
A_j	1.20	0.92
B_j	0.0065	0.0048
C_j	0.000279	0.000125

The MPC controller parameters $N_p = 20$, $N_c = 15$, and weight coefficients are shown in Table II.

TABLE II: Parameters of weight coefficients

K^l_v	K^l_{in}	K^l_u	K^f_v	K^f_D	K^f_{in}	K^f_u
1×10^9	1×10^2	1×10^4	1×10^7	1×10^3	1×10^2	100

The maximum traction and braking forces of HXD2 locomotives (both measured in kN) are given by the following equations,

$$u^l_{max}(v) = \begin{cases} 760, & v \le 5 \\ 779.05 - 3.81v, & 5 < v \le 65 \\ 34541/v, & 65 < v \le 120 \\ 0, & v > 120 \end{cases} \tag{13}$$

and

$$u^l_{min}(v) = \begin{cases} 153.3v, & v \le 3 \\ 460, & 5 < v \le 65 \\ 34500/v, & 65 < v \le 120 \\ 0, & v > 120 \end{cases} \tag{14}$$

where v is in km/h. v^{lim} is set to $35m/s$.

we define two indicators $\overline{f^{in}_i}$ and E to measure the in-train force and total energy consumption, respectively. The mean value of the in-train force $f^{in}_{i,j}$ of ith HHT within the operation period is defined as

$$\overline{f^{in}_i} = \frac{1}{N_t}\int_0^{N_t} \left|f^{in}_{i,j}(t)\right| dt, \quad i = 1, 2, ..., m; j = 1, 2, ..., n-1. \tag{15}$$

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A smaller value of $\overline{f_j^{in}}$ indicates the improved safety and less cost of train maintenance, and $f_{max}^{in} = 2000kN$. We also define an indicator for the total energy consumption during operation as

$$E_i = \sum_{j=1}^{n} \int_0^{N_t} |u_{i,j} v_{i,j}| \, dt, \quad i = 1, 2, ..., m; j = 1, 2, ..., n. \tag{16}$$

The variables $u_{i,j}$ and $v_{i,j}$ represent the control force and speed of jth vehicle in ith HHT, respectively.

The simulation results are categorized into two cases: A. Case1: without disturbance, B. Case 2: HHTs 2 and 4 with input disturbances. Five HHTs with the same formation are selected as the virtual coupling group, the minimum tracking interval D_{min} and the desired interval D_{des} between the HHTs are 500 meters.

A. Case 1: Virtually Coupled HHTs Without Disturbance

In order to show the operation state of virtual coupling HHTs more clearly, this paper takes the mean speeds of all the mass points of each HHT, together with the mean speed errors and the tracking interval errors, which are shown in Fig. 5.

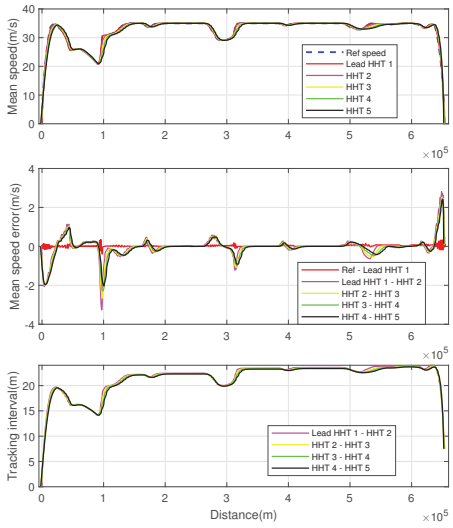


Fig. 5: Trajectories of mean speeds, mean speed errors and tracking interval errors in case 1.

As illustrated in Fig. 5, the DMPC controller designed in this paper enables the virtual coupling HHTs to demonstrate excellent performance in speed tracking and interval maintenance.

Fig. 6 illustrates the traction and braking forces of locomotives, where the solid line indicates the first locomotive in each HHT and the dashed line indicates the second. The in-train force mean values of each HHT are the average value of in-train forces per coupler. Although there were some moments of

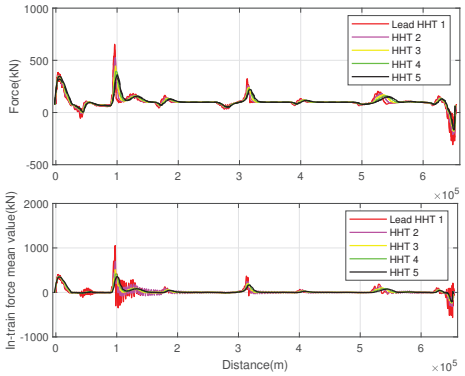


Fig. 6: Trajectories of locomotive forces and in-train force mean values in case 1.

significant fluctuation, the in-train foeces were always within the safe range.

B. Case 2: Virtually Coupled HHTs With Input Disturbances in HHTs 2 and 4.

In this subsection, random disturbances is introduced into the locomotive's control inputs in HHT 2 and HHT 4 during time interval $500 \sim 1000$ seconds, the values of which are randomly taken from $[-10, 10]kN$. The simulation results are illustrated as Figs. 7 and 8.

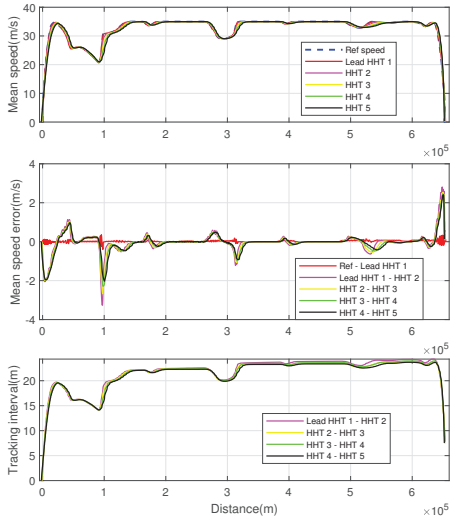


Fig. 7: Trajectories of mean speeds, mean speed errors and tracking interval errors in case 3.

The fluctuations in the in-train force curves for HHT 2 and HHT 4 are more pronounced, but the speed tracking errors and the tracking interval errors remain almost unchanged.

The data of the simulation results are shown in table III, from which it can be seen that the presence of random

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TABLE III: The performance indicators of virtual coupling HHTs under different scenarios.

Parameters	Case	Leader HHT 1	HHT 2	HHT3	HHT 4	HHT 5
Mean values of	Case1	0.040	0.346	0.342	0.339	0.337
speed error (m/s)	Case2	0.040	**0.348**	0.343	**0.336**	0.337
Mean values of	Case1	0	19.442	19.318	19.247	19.179
Tracking interval error (m)	Case2	0	**19.628**	19.325	**19.181**	19.185
Mean values of	Case1	52.140	48.254	42.157	40.742	40.035
in-train force (kN)	Case2	52.140	**60.137**	42.958	**52.682**	42.331
Energy consumption	Case1	18.50	18.40	18.32	18.26	18.22
(GJ)	Case2	18.50	**18.39**	18.35	**18.28**	18.26

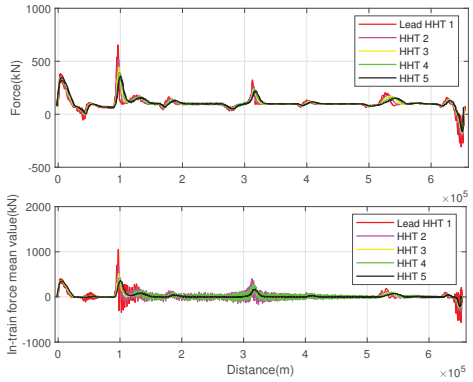


Fig. 8: Trajectories of locomotive forces and in-train force mean values in case 3.

disturbances has almost no effect on the speed tracking errors and interval errors, due to the fact that the speed tracking error mean value remains almost unchanged. The most variable metric is the in-train force mean values, but it is still within safe limits and has almost no effect on the whole virtual coupling HHTs.

V. CONCLUSION

In order to enhance the capacity and operational efficiency of HHTs, we have proposed a virtual coupling strategy for HHTs based on a multi-mass-point model in this paper. DMPC controllers are designed for a group of virtually coupled HHTs with the objectives of reaching the same speed while minimizing tracking intervals, in-train forces and energy consumption. The multi-objective receding horizon optimization problem is transformed into a nonlinear programming form to solve. The effectiveness of the proposed method is verified by numerical simulations, which demonstrate its excellent performance in speed tracking, interval maintenance and minimizing in-train force, even in the presence of random disturbances. It will be our future work to further consider the highly nonlinear operation process of heavy-haul trains, and to propose a virtual coupling strategy that is more suitable for the cooperative operation of HHT group with long formations.

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